

# Zotero Evidence: AI-assisted systematic reviews with traceable evidence

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DOI: [DOIunavailable](#)

## Software

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Submitted: N/A

Published: N/A

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## Summary

Zotero Evidence is an open-source plugin that turns the reference manager Zotero into an end-to-end workbench for systematic reviews, keeping every conclusion traceable back to its supporting evidence. Working entirely inside a user's Zotero library, it guides a review through the full pipeline: importing search results, title/abstract and full-text screening, codebook-based data extraction, thematic synthesis, and reporting exports. Each stage can be assisted by a user-configured, OpenAI-compatible large language model (LLM), yet the reviewer remains the decision maker — every AI output is an advisory suggestion that a human must confirm. Traceability is enforced at the point of extraction: each confirmed value is bound to the exact quote and highlight in the source PDF that supports it, so any datum in the final synthesis can be followed back to its origin. Projects are stored as a fixed collection tree in a single Zotero library and can be exported as a portable archive for backup or sharing, or as CSV files for PRISMA reporting (Page et al., 2021), screening logs, coded data, and synthesis output.

## Statement of need

Systematic reviews are foundational to evidence-based research but are labor-intensive: reviewers must screen large numbers of records, assess full texts against explicit eligibility criteria, and extract data reproducibly. Existing tools address only parts of this workload. Rayyan (Ouzzani et al., 2016) and ASReview (Schoot et al., 2021) accelerate abstract screening, the latter through active-learning prioritization, and several commercial platforms support review management. However, such tools are typically standalone web services, separate from the reference manager where researchers already collect and read their literature, and most stop at screening rather than continuing into the equally demanding steps of quote-backed extraction and synthesis.

Zotero Evidence addresses three gaps. First, it embeds the whole workflow in Zotero, so items, PDFs, annotations, and review decisions live together in the user's own library and sync through existing Zotero infrastructure, including group libraries for teams. Second, it extends AI assistance across the entire pipeline — title/abstract judgments, per-criterion full-text checks, codebook extraction, and theme mining — while keeping the reviewer in control: AI results are never final on their own, and the plugin marks an explicit human confirmation point at each stage. Third, it makes the review auditable: extracted values are tied to verbatim quotes and PDF highlights, screening criteria are versioned, and agreement between AI and human, or between two human reviewers, can be quantified with Cohen's kappa (Cohen, 1960) for quality control.

By combining reference management, pipeline-wide AI assistance, and human-confirmed traceability in a single free and open-source tool, Zotero Evidence gives researchers, students, librarians, and review teams a transparent, reproducible alternative to fragmented or closed review workflows.

## Key features

- **Import and deduplication.** Import RIS, BibTeX, MEDLINE, or PubMed XML via Zotero's translators, with DOI-first duplicate detection and a year/author/title fallback.
- **Two-stage screening.** AI suggestions for title/abstract inclusion and per-criterion full-text checks, each requiring human confirmation, with results routed into a fixed collection structure.
- **Codebook extraction.** User-defined variables (categorical, numeric, or text) extracted from full text, each confirmed value linked to a supporting quote and PDF highlight.
- **Consistency.** Human–AI and human–human agreement reported as Cohen's kappa, including a sampled dual-reviewer reconciliation workflow.
- **Synthesis and export.** One-variable theme mining over confirmed evidence, and CSV exports for PRISMA flow counts, screening logs, coded data, and synthesis output.
- **Privacy by design.** Data stays in the user's Zotero library; only the minimal prompt content needed for a chosen AI feature is sent to the user-configured provider, and the plugin includes no telemetry.

## Acknowledgements

Zotero Evidence is built on the Zotero platform (Corporation for Digital Scholarship, n.d.) and the `zotero-plugin-toolkit` and `zotero-plugin-scaffold` projects.

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